ASSIGNMENT-1

Introduction to Machine Learning

Course: MCA

Subject: Machine Learning

Question 1 (5 Marks)

Define Supervised Learning. Explain the key difference between supervised learning and unsupervised learning with one example of each.

Supervised Learning is a type of Machine Learning in which a model learns from labelled data containing input features and their correct outputs. It uses these examples to predict outputs for new, unseen data.

Difference between Supervised and Unsupervised Learning:

Supervised Learning	Unsupervised Learning
Uses labelled data.	Uses unlabelled data.
Learns to predict known outputs.	Identifies hidden patterns or groups.
Example: Predicting whether an email is spam.	Example: Grouping customers by purchasing behaviour.

Conclusion: Supervised learning predicts outcomes using labelled examples, whereas unsupervised learning discovers patterns without predefined labels.

Question 2 (5 Marks)

What are the two main types of supervised learning problems? Clearly distinguish between them and give one real-world example for each type.

The two main types of supervised learning are:

1. Classification: Classification predicts a category or class from input data. Its output is a discrete label.

- Example: Classifying a transaction as fraudulent or legitimate.
- Suitable algorithm: Decision Tree.

2. Regression: Regression predicts a continuous numerical value based on input features.

- Example: Predicting the price of a house using its area and location.
- Suitable algorithm: Linear Regression.

Basis	Classification	Regression
Output	Category or class	Numerical value

Example	Spam detection	House price prediction
Algorithms	Logistic Regression, Decision Tree	Linear Regression, Random Forest Regression

Conclusion: Classification predicts categories, while regression predicts numerical values.

Question 3 (5 Marks)

Explain the concept of Linear Regression. Write the equation of a simple linear regression model and define its terms. What is the role of the residual/error term?

Linear Regression is a supervised learning algorithm used to predict a continuous numerical value by identifying a linear relationship between independent and dependent variables.

The equation of a simple linear regression model is:

$$ Y = \beta_0 + \beta_1X + \varepsilon $$

Where:

- Y — Dependent Variable: The value to be predicted, such as house price.
- X — Independent Variable: The input used for prediction, such as house area.
- β_0 — Intercept: The predicted value of Y when $X=0$.
- β_1 — Slope: The expected change in Y for a one-unit increase in X.
- ε — Error Term: Represents the influence of factors not explained by the linear model.

Example: A model predicts house prices based on house area. Generally, a larger house may have a higher predicted price.

The residual is the difference between the actual and predicted value:

$$ e_i = Y_i-\hat{Y}_i $$

The error term represents unexplained variation in the statistical model, while the residual is its estimated counterpart for an observed data point.

Conclusion: Linear Regression models linear relationships and helps estimate continuous numerical values.

Question 4 (5 Marks)

How does Logistic Regression differ from Linear Regression? Write the logistic function and explain why it is suitable for binary classification. What is the decision boundary?

Linear Regression predicts continuous numerical values, whereas Logistic Regression estimates the probability of a class, commonly for binary classification.

Logistic (Sigmoid) Function

$$ P(Y=1\mid X)=\frac{1}{1+e^{-z}} $$

Where:

$$ z=\beta_0+\beta_1X $$

The sigmoid function converts the linear score into a probability between 0 and 1.

Why Logistic Regression Is Suitable

- It estimates the probability of belonging to a class.
- Its output lies between 0 and 1.
- A threshold, commonly 0.5, can be used to assign a class label.

Example: If the predicted probability that an email is spam is 0.8, the model classifies it as spam using a threshold of 0.5.

Decision Boundary

The decision boundary is the point or line that separates the predicted classes. With a threshold of 0.5, the boundary occurs where:

$$ z=0 $$

Conclusion: Logistic Regression is suitable for binary classification because it converts input features into class probabilities and uses a decision boundary to separate classes.

Question 5 (5 Marks)

Describe how a Decision Tree works for classification. Explain Information Gain and Gini Impurity. Why can decision trees overfit, and how can this be reduced?

A Decision Tree is a supervised learning algorithm that classifies data through a sequence of conditions. It contains a root node, decision nodes, branches, and leaf nodes.

Working

1. The algorithm selects a feature for splitting the data.
2. It creates branches based on the feature's values or conditions.
3. It repeats the process on the resulting groups.
4. The leaf node provides the final predicted class.

Information Gain

Information Gain measures how much a split reduces uncertainty or entropy in the data.

$$ IG=H(\text{Parent})- \sum_j\frac{N_j}{N}H(\text{Child}_j) $$

A split with higher Information Gain is generally preferred.

Gini Impurity

Gini Impurity measures the likelihood of incorrectly classifying a randomly selected sample if its class is assigned according to the class distribution of the node.

$$ Gini=1-\sum_{i=1}^{k}p_i^2 $$

A lower Gini value indicates a purer node.

Overfitting and Prevention

Decision Trees can overfit when they grow too deep and learn noise or specific details from the training data.

Overfitting can be reduced by:

- Limiting the maximum tree depth.
- Setting a minimum number of samples per leaf.
- Pruning unnecessary branches.
- Using cross-validation to select suitable parameters.

Conclusion: Decision Trees are easy to interpret, but controlling their complexity is important for good performance on unseen data.

Question 6 (5 Marks)

What are Ensemble Methods? Differentiate between Bagging and Boosting with suitable examples. Why do ensemble methods generally perform better than a single model?

Ensemble Methods combine predictions from multiple models to improve predictive performance, stability, or generalization.

Bagging vs. Boosting

Basis	Bagging	Boosting
Working	Trains models independently, often in parallel.	Trains models sequentially, with later models focusing on previous errors.
Main purpose	Mainly reduces variance.	Often reduces bias and improves prediction accuracy.
Data	Uses bootstrap samples of training data.	Uses reweighting or error-based updates, depending on the algorithm.
Example	Random Forest	AdaBoost, Gradient Boosting

Why Ensemble Methods Perform Better

1. They combine information from multiple models.
2. They can reduce errors made by individual models.
3. Bagging can improve stability and reduce variance.
4. Boosting can progressively improve prediction performance.

Example: Random Forest combines predictions from many decision trees rather than relying on a single tree.

Conclusion: Ensemble methods often provide more accurate and robust predictions than individual models, although their performance depends on the data and algorithm used.

Question 7 (5 Marks)

Explain the working principle of Support Vector Machines (SVM) for binary classification. What are support vectors and the margin? Briefly describe the role of the kernel trick.

Support Vector Machine (SVM) is a supervised learning algorithm used for classification. It finds a decision boundary, called a hyperplane, that separates two classes while maximizing the margin between them.

Working Principle

1. SVM identifies a hyperplane that separates the classes.
2. It searches for the hyperplane with the maximum margin.
3. New data points are classified according to which side of the hyperplane they fall on.

Important Concepts

Support Vectors: These are the training data points closest to the decision boundary. They play a key role in determining the position of the hyperplane.

Margin: The margin is the distance between the decision boundary and the nearest data points from the classes. A larger margin can improve generalization.

Kernel Trick: When data cannot be separated using a straight line or flat hyperplane in the original feature space, a kernel function allows SVM to model nonlinear decision boundaries without explicitly calculating the transformed features.

Common kernels include Linear, Polynomial, and Radial Basis Function (RBF).

Example: SVM can classify emails as spam or non-spam based on their features.

Conclusion: SVM aims to separate classes with a maximum-margin boundary and can handle nonlinear classification using suitable kernels.

Question 8 (5 Marks)

List and briefly explain any four evaluation metrics used for classification. When is Accuracy not a good metric, and which metric would you prefer instead?

Classification metrics measure how well a model predicts class labels.

1. Accuracy

The proportion of all predictions that are correct.

$$ Accuracy=\frac{TP+TN}{TP+TN+FP+FN} $$

2. Precision

The proportion of predicted positive cases that are actually positive.

$$ Precision=\frac{TP}{TP+FP} $$

3. Recall

The proportion of actual positive cases correctly identified.

$$ Recall=\frac{TP}{TP+FN} $$

4. F1-Score

The harmonic mean of precision and recall.

$$ F1=2\frac{Precision\times Recall}{Precision+Recall} $$

Here, TP means true positives, TN means true negatives, FP means false positives, and FN means false negatives.

When Is Accuracy Misleading?

Accuracy may be misleading when the dataset is highly imbalanced.

Example: In a dataset of 1,000 transactions, 990 are legitimate and 10 are fraudulent. A model that predicts every transaction as legitimate achieves 99% accuracy but detects no fraud.

In this case, Recall is important when detecting as many fraudulent transactions as possible. Precision should also be considered to control false alerts, while F1-score can balance precision and recall.

Conclusion: The best metric depends on the problem, class distribution, and cost of prediction errors.

Question 9 (5 Marks)

For regression problems, explain MSE, RMSE, MAE, and R-squared. Which metric is more sensitive to outliers and why?

Regression evaluation metrics measure the difference between actual and predicted numerical values.

(a) Mean Squared Error (MSE)

MSE is the average of the squared prediction errors.

$$ MSE=\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat y_i)^2 $$

Larger errors receive greater penalties because errors are squared.

(b) Root Mean Squared Error (RMSE)

RMSE is the square root of MSE.

$$ RMSE=\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat y_i)^2} $$

It is expressed in the same units as the target variable.

(c) Mean Absolute Error (MAE)

MAE is the average absolute difference between actual and predicted values.

$$ MAE=\frac{1}{n}\sum_{i=1}^{n}|y_i-\hat y_i| $$

It is easy to interpret and penalizes errors proportionally to their absolute size.

(d) R-squared (Coefficient of Determination)

R-squared measures how much of the variation in the target variable is explained by the regression model.

$$ R^2=1-\frac{\sum_i(y_i-\hat y_i)^2}{\sum_i(y_i-\bar y)^2} $$

A value closer to 1 generally indicates a better fit compared with predicting the target mean, although R-squared alone does not establish that a model is good.

Which Metric Is More Sensitive to Outliers?

MSE is more sensitive to outliers because it squares prediction errors. RMSE is also strongly affected because it is derived from MSE. MAE is generally less sensitive because it uses absolute errors rather than squared errors.

Conclusion: MSE and RMSE penalize large errors heavily, MAE is more robust to extreme errors, and R-squared measures the explained variation.

Question 10 (5 Marks)

Discuss any four real-world applications of Supervised Learning. For each application, mention the type of problem and a suitable algorithm.

Supervised Learning is used to predict outputs from labelled historical data. Four important applications are:

Application	Type	Suitable Algorithm
Email spam detection	Classification	Naive Bayes
House price prediction	Regression	Linear Regression
Fraud detection	Classification	Random Forest
Sales forecasting	Regression	Gradient Boosting Regression

1. Email Spam Detection

A model learns from emails labelled as spam or non-spam and classifies new emails. Naive Bayes is a suitable algorithm for this task.

2. House Price Prediction

A model uses historical house prices, area, location, and other features to estimate the price of a property. Linear Regression can model relationships between these features and price.

3. Fraud Detection

A model learns from labelled transaction records to identify potentially fraudulent transactions. Random Forest can capture complex patterns in transaction data.

4. Sales Forecasting

A model uses historical sales and relevant factors, such as prices and promotions, to predict future sales values. Gradient Boosting Regression is one suitable approach.